

```
mysql> notee
mysql> /* System informatic functions
/*> It genrates the metadata of the system
/*> curdate() = current date
/*> curtime() = current time
/*> now () = current date and time */
```

```
mysql> select curdate();
```

```
+-----+
```

```
| curdate() |
```

```
+-----+
```

```
| 2026-09-14 |
```

```
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select curtime();
```

```
+-----+
```

```
| curtime() |
```

```
+-----+
```

```
| 14:14:59 |
```

```
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select now();
```

```
+-----+
```

```
| now() |
```

```
+-----+
```

```
| 2026-09-14 14:15:19 |
```

```
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select database();
```

```
+-----+
```

```
| database() |
```

```
+-----+
```

```
| cg_mysql |
```

```
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select version();
```

```
+-----+
```

```
| version() |
```

```
+-----+
```

```
| 8.0.41 |
```

```
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select sysdate();
```

```
+-----+
```

```
| sysdate() |
```

```
+-----+
```

```
| 2026-09-14 14:16:04 |
```

```
+-----+
1 row in set (0.00 sec)
```

```
mysql> /* Aggregate functions
```

```
/*> 1.count()
```

```
/*> 2.min()
```

```
/*> 3.max()
```

```
/*> 4.avg()
```

```
/*> 5.sum
```

```
/*> They are applicable only for numeric datatype */
```

```
mysql> select * from employees;
```

emp_id	emp_name	department	salary	experience	age
101	Ravi	IT	65000.00	5	28
102	Priya	HR	45000.00	2	24
103	Amit	Finance	75000.00	8	35
104	Sneha	IT	55000.00	3	27
105	Rahul	Sales	40000.00	1	22
106	Kiran	Finance	90000.00	12	40
107	Anjali	HR	52000.00	4	29
108	Suresh	Sales	68000.00	7	33

```
8 rows in set (0.03 sec)
```

```
mysql> select count(*) from employees;
```

```
+-----+
| count(*) |
```

```
+-----+
|      8 |
```

```
+-----+
1 row in set (0.01 sec)
```

```
mysql> -- count function does not count null value, it only counts rows with values
```

```
mysql> select max(experience) from employees;
```

```
+-----+
| max(experience) |
```

```
+-----+
|             12 |
```

```
+-----+
1 row in set (0.01 sec)
```

```
mysql> select min(experience) as low_experience from employees;
```

```
+-----+
| low_experience |
```

```
+-----+
|             1 |
```

```
+-----+
1 row in set (0.00 sec)
```

```
mysql> select max(salary) as max_salary from employees;
```

```
+-----+
| max_salary |
+-----+
|  90000.00 |
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select max(experience) as high_experience from employees;
```

```
+-----+
| high_experience |
+-----+
|              12 |
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select min(salary) as min_salary from employees;
```

```
+-----+
| min_salary |
+-----+
|  40000.00 |
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select avg(salary) as avg_salary from employees;
```

```
+-----+
| avg_salary |
+-----+
| 61250.000000 |
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select sum(salary) as tota_salary from employees;
```

```
+-----+
| tota_salary |
+-----+
| 490000.00 |
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> /* Clauses
```

```
/*> -> In mysql clauses are used to control how data is filtered,
grouped,sorted or modified in SQLqueries.
```

```
/*> -> They make SQL Statements more meaningful and powerful.
```

```
/*> WHERE Clause: It is used filters records based on specified conditions
where clause can be used as comparison operators,logical operators and etc.
```

```
/*> syntax: select column_name from table_name where condition;
```

```
/*>
```

```
/*> Group by: It is used to group rows having the same values in one or more
columns, it is commonly used with aggregate functions such as
min(),max(),sum(),count(),avg().
```

```

/*> syntax:select column_name/aggregate_function(column_name) from table_name
group by column_name;
/*> */
mysql> /* having clause: It is used to filter grouped records after group by
clause. It is similar to where but it works on grouped instead of individual rows.
/*> syntax: select column_name/aggregate_function(column_name) from table_name
group by column_name having condition;
/*>
/*> order by clause: It is used to sort the result set in ascending(ASC) or
descending(DESC) order
/*> syntax: select column_name from table_name order by column_name;
/*> */

```

```

mysql>
mysql> show tables;
+-----+
| Tables_in_cg_mysql |
+-----+
| backup_emp          |
| company_staff       |
| departments         |
| doctors             |
| emp                 |
| employees            |
| orders              |
| products            |
| projects            |
| studentscores       |
+-----+
10 rows in set (0.03 sec)

```

```

mysql> CREATE TABLE STUDENTS (
->     StudentID INT AUTO_INCREMENT PRIMARY KEY,
->     Name VARCHAR(50),
->     Age INT,
->     Department VARCHAR(30),
->     Marks INT,
->     City VARCHAR(30)
-> );
Query OK, 0 rows affected (0.04 sec)

```

```

mysql> INSERT INTO STUDENTS (Name, Age, Department, Marks, City) VALUES
-> ('Anil', 20, 'CSE', 85, 'Hyderabad'),
-> ('Sneha', 22, 'ECE', 75, 'Mumbai'),
-> ('Ravi', 20, 'CSE', NULL, 'Hyderabad'),
-> ('Divya', 23, 'MECH', 65, NULL),
-> ('Kiran', 22, 'ECE', 75, 'Pune'),
-> ('Varun', 21, 'CSE', 92, 'Chennai'),
-> ('Priya', 20, 'EEE', 88, 'Delhi'),
-> ('John', 22, 'MECH', 55, 'Mumbai'),
-> ('Rekha', 21, 'ECE', 75, 'Pune'),
-> ('Ramesh', 23, 'EEE', NULL, 'Delhi');

```

Query OK, 10 rows affected (0.02 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> select * from students;

StudentID	Name	Age	Department	Marks	City
1	Anil	20	CSE	85	Hyderabad
2	Sneha	22	ECE	75	Mumbai
3	Ravi	20	CSE	NULL	Hyderabad
4	Divya	23	MECH	65	NULL
5	Kiran	22	ECE	75	Pune
6	Varun	21	CSE	92	Chennai
7	Priya	20	EEE	88	Delhi
8	John	22	MECH	55	Mumbai
9	Rekha	21	ECE	75	Pune
10	Ramesh	23	EEE	NULL	Delhi

10 rows in set (0.00 sec)

mysql> select * from students where department='CSE';

StudentID	Name	Age	Department	Marks	City
1	Anil	20	CSE	85	Hyderabad
3	Ravi	20	CSE	NULL	Hyderabad
6	Varun	21	CSE	92	Chennai

3 rows in set (0.00 sec)

mysql> select * from students where Marks>80;

StudentID	Name	Age	Department	Marks	City
1	Anil	20	CSE	85	Hyderabad
6	Varun	21	CSE	92	Chennai
7	Priya	20	EEE	88	Delhi

3 rows in set (0.00 sec)

mysql> select * from students where Age=22;

StudentID	Name	Age	Department	Marks	City
2	Sneha	22	ECE	75	Mumbai
5	Kiran	22	ECE	75	Pune
8	John	22	MECH	55	Mumbai

3 rows in set (0.00 sec)

mysql> select * from students where city='Mumbai';

```
+-----+-----+-----+-----+-----+-----+
| StudentID | Name | Age | Department | Marks | City |
+-----+-----+-----+-----+-----+-----+
|          2 | Sneha | 22 | ECE          | 75 | Mumbai |
|          8 | John  | 22 | MECH         | 55 | Mumbai |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> select * from students where marks<70;
```

```
+-----+-----+-----+-----+-----+-----+
| StudentID | Name | Age | Department | Marks | City |
+-----+-----+-----+-----+-----+-----+
|          4 | Divya | 23 | MECH         | 65 | NULL |
|          8 | John  | 22 | MECH         | 55 | Mumbai |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> select * from students where marks between 70 and 90;
```

```
+-----+-----+-----+-----+-----+-----+
| StudentID | Name | Age | Department | Marks | City |
+-----+-----+-----+-----+-----+-----+
|          1 | Anil  | 20 | CSE          | 85 | Hyderabad |
|          2 | Sneha | 22 | ECE          | 75 | Mumbai |
|          5 | Kiran | 22 | ECE          | 75 | Pune |
|          7 | Priya | 20 | EEE          | 88 | Delhi |
|          9 | Rekha | 21 | ECE          | 75 | Pune |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.01 sec)
```

```
mysql> select * from students where Age<>20;
```

```
+-----+-----+-----+-----+-----+-----+
| StudentID | Name | Age | Department | Marks | City |
+-----+-----+-----+-----+-----+-----+
|          2 | Sneha | 22 | ECE          | 75 | Mumbai |
|          4 | Divya | 23 | MECH         | 65 | NULL |
|          5 | Kiran | 22 | ECE          | 75 | Pune |
|          6 | Varun | 21 | CSE          | 92 | Chennai |
|          8 | John  | 22 | MECH         | 55 | Mumbai |
|          9 | Rekha | 21 | ECE          | 75 | Pune |
|         10 | Ramesh | 23 | EEE          | NULL | Delhi |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)
```

```
mysql> select * from students where department='CSE'and marks>80;
```

```
+-----+-----+-----+-----+-----+-----+
| StudentID | Name | Age | Department | Marks | City |
+-----+-----+-----+-----+-----+-----+
|          1 | Anil  | 20 | CSE          | 85 | Hyderabad |
|          6 | Varun | 21 | CSE          | 92 | Chennai |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> select * from students where department='ECE' or department='EEE';
```

StudentID	Name	Age	Department	Marks	City
2	Sneha	22	ECE	75	Mumbai
5	Kiran	22	ECE	75	Pune
7	Priya	20	EEE	88	Delhi
9	Rekha	21	ECE	75	Pune
10	Ramesh	23	EEE	NULL	Delhi

5 rows in set (0.00 sec)

```
mysql> select * from students where city != 'Delhi';
```

StudentID	Name	Age	Department	Marks	City
1	Anil	20	CSE	85	Hyderabad
2	Sneha	22	ECE	75	Mumbai
3	Ravi	20	CSE	NULL	Hyderabad
5	Kiran	22	ECE	75	Pune
6	Varun	21	CSE	92	Chennai
8	John	22	MECH	55	Mumbai
9	Rekha	21	ECE	75	Pune

7 rows in set (0.00 sec)

```
mysql> -- Group by clause
```

```
mysql> select Department,count(*) as total_students from students group by Department;
```

Department	total_students
CSE	3
ECE	3
MECH	2
EEE	2

4 rows in set (0.00 sec)

```
mysql> select Department,AVG(marks) as Avg_marks from students group by Department;
```

Department	Avg_marks
CSE	88.5000
ECE	75.0000
MECH	60.0000
EEE	88.0000

4 rows in set (0.00 sec)

```
mysql> select Department,Max(marks) as Highest_marks from students group by Department;
```

Department	Highest_marks
CSE	92
ECE	75
MECH	65
EEE	88

4 rows in set (0.00 sec)

```
mysql> select Department,Min(marks) as lowest_marks from students group by Department;
```

Department	lowest_marks
CSE	85
ECE	75
MECH	55
EEE	88

4 rows in set (0.00 sec)

```
mysql> select Department,sum(marks) as total_marks from students group by Department;
```

Department	total_marks
CSE	177
ECE	225
MECH	120
EEE	88

4 rows in set (0.00 sec)

```
mysql> select city,count(*) as total_students from students group by city;
```

city	total_students
Hyderabad	2
Mumbai	2
NULL	1
Pune	2
Chennai	1
Delhi	2

6 rows in set (0.00 sec)

```
mysql> select Department,AVG(age) as Avg_age from students group by Department;
```

```

+-----+-----+
| Department | Avg_age |
+-----+-----+
| CSE        | 20.3333 |
| ECE        | 21.6667 |
| MECH       | 22.5000 |
| EEE        | 21.5000 |
+-----+-----+
4 rows in set (0.00 sec)

```

```
mysql> select city,sum(marks) as total_students from students group by city;
```

```

+-----+-----+
| city      | total_students |
+-----+-----+
| Hyderabad | 85             |
| Mumbai    | 130            |
| NULL      | 65             |
| Pune      | 150            |
| Chennai   | 92             |
| Delhi     | 88             |
+-----+-----+
6 rows in set (0.00 sec)

```

```
mysql> select city,max(marks) as highest_students from students group by city;
```

```

+-----+-----+
| city      | highest_students |
+-----+-----+
| Hyderabad | 85              |
| Mumbai    | 75              |
| NULL      | 65              |
| Pune      | 75              |
| Chennai   | 92              |
| Delhi     | 88              |
+-----+-----+
6 rows in set (0.00 sec)

```

```
mysql> select city,min(marks) as lowest_students from students group by city;
```

```

+-----+-----+
| city      | lowest_students |
+-----+-----+
| Hyderabad | 85              |
| Mumbai    | 55              |
| NULL      | 65              |
| Pune      | 75              |
| Chennai   | 92              |
| Delhi     | 88              |
+-----+-----+
6 rows in set (0.00 sec)

```

```
mysql> select age,count(*) as total_students from students group by age;
```

```

+-----+-----+

```

age	total_students
20	3
22	3
23	2
21	2

4 rows in set (0.00 sec)

mysql> -- Having clause

mysql> select Department,count(*) as total_students from students group by Department having count(*)>2;

Department	total_students
CSE	3
ECE	3

2 rows in set (0.00 sec)

mysql> select Department,avg(marks) as avg_marks from students group by Department having avg(marks)>75;

Department	avg_marks
CSE	88.5000
EEE	88.0000

2 rows in set (0.00 sec)

mysql> select city,count(*) as total_students from students group by city having count(*)>1;

city	total_students
Hyderabad	2
Mumbai	2
Pune	2
Delhi	2

4 rows in set (0.00 sec)

mysql> select Department,sum(marks) as total_marks from students group by Department having sum(marks)>150;

Department	total_marks
CSE	177
ECE	225

2 rows in set (0.00 sec)

=

```
mysql> select age,count(*) as total_students from students group by age having count(*)>2;
```

age	total_students
20	3
22	3

2 rows in set (0.00 sec)

```
mysql> select city,avg(marks) as avg_marks from students group by city having avg(marks)>70;
```

city	avg_marks
Hyderabad	85.0000
Pune	75.0000
Chennai	92.0000
Delhi	88.0000

4 rows in set (0.00 sec)

```
mysql> select Department,max(marks) as highest_marks from students group by Department having max(marks)>90;
```

Department	highest_marks
CSE	92

1 row in set (0.00 sec)

```
mysql> select Department,min(marks) as lowest_marks from students group by Department having min(marks)>60;
```

Department	lowest_marks
CSE	85
ECE	75
EEE	88

3 rows in set (0.00 sec)

```
mysql> select city,sum(marks) as total_students from students group by city having sum(marks)>100;
```

city	total_students
Mumbai	130
Pune	150

```
+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> select Department,count(*) as total_students from students group by
Department having count(*)=2;
```

```
+-----+-----+
| Department | total_students |
+-----+-----+
| MECH       |                2 |
| EEE        |                2 |
+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> -- order by clause
mysql> select * from students order by name;
```

```
+-----+-----+-----+-----+-----+-----+
| StudentID | Name   | Age | Department | Marks | City       |
+-----+-----+-----+-----+-----+-----+
|          1 | Anil   | 20  | CSE        | 85    | Hyderabad |
|          4 | Divya  | 23  | MECH       | 65    | NULL      |
|          8 | John   | 22  | MECH       | 55    | Mumbai    |
|          5 | Kiran  | 22  | ECE        | 75    | Pune      |
|          7 | Priya  | 20  | EEE        | 88    | Delhi     |
|         10 | Ramesh | 23  | EEE        | NULL  | Delhi     |
|          3 | Ravi   | 20  | CSE        | NULL  | Hyderabad |
|          9 | Rekha  | 21  | ECE        | 75    | Pune      |
|          2 | Sneha  | 22  | ECE        | 75    | Mumbai    |
|          6 | Varun  | 21  | CSE        | 92    | Chennai   |
+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> select * from students order by marks desc;
```

```
+-----+-----+-----+-----+-----+-----+
| StudentID | Name   | Age | Department | Marks | City       |
+-----+-----+-----+-----+-----+-----+
|          6 | Varun  | 21  | CSE        | 92    | Chennai   |
|          7 | Priya  | 20  | EEE        | 88    | Delhi     |
|          1 | Anil   | 20  | CSE        | 85    | Hyderabad |
|          2 | Sneha  | 22  | ECE        | 75    | Mumbai    |
|          5 | Kiran  | 22  | ECE        | 75    | Pune      |
|          9 | Rekha  | 21  | ECE        | 75    | Pune      |
|          4 | Divya  | 23  | MECH       | 65    | NULL      |
|          8 | John   | 22  | MECH       | 55    | Mumbai    |
|          3 | Ravi   | 20  | CSE        | NULL  | Hyderabad |
|         10 | Ramesh | 23  | EEE        | NULL  | Delhi     |
+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> select * from students order by age;
```

```
+-----+-----+-----+-----+-----+-----+
| StudentID | Name   | Age | Department | Marks | City       |
+-----+-----+-----+-----+-----+-----+
```

1	Anil	20	CSE	85	Hyderabad
3	Ravi	20	CSE	NULL	Hyderabad
7	Priya	20	EEE	88	Delhi
6	Varun	21	CSE	92	Chennai
9	Rekha	21	ECE	75	Pune
2	Sneha	22	ECE	75	Mumbai
5	Kiran	22	ECE	75	Pune
8	John	22	MECH	55	Mumbai
4	Divya	23	MECH	65	NULL
10	Ramesh	23	EEE	NULL	Delhi

10 rows in set (0.00 sec)

mysql> select * from students order by department;

StudentID	Name	Age	Department	Marks	City
1	Anil	20	CSE	85	Hyderabad
3	Ravi	20	CSE	NULL	Hyderabad
6	Varun	21	CSE	92	Chennai
2	Sneha	22	ECE	75	Mumbai
5	Kiran	22	ECE	75	Pune
9	Rekha	21	ECE	75	Pune
7	Priya	20	EEE	88	Delhi
10	Ramesh	23	EEE	NULL	Delhi
4	Divya	23	MECH	65	NULL
8	John	22	MECH	55	Mumbai

10 rows in set (0.00 sec)

mysql> select * from students order by city;

StudentID	Name	Age	Department	Marks	City
4	Divya	23	MECH	65	NULL
6	Varun	21	CSE	92	Chennai
7	Priya	20	EEE	88	Delhi
10	Ramesh	23	EEE	NULL	Delhi
1	Anil	20	CSE	85	Hyderabad
3	Ravi	20	CSE	NULL	Hyderabad
2	Sneha	22	ECE	75	Mumbai
8	John	22	MECH	55	Mumbai
5	Kiran	22	ECE	75	Pune
9	Rekha	21	ECE	75	Pune

10 rows in set (0.00 sec)

mysql> select * from students order by department,marks desc;

StudentID	Name	Age	Department	Marks	City
-----------	------	-----	------------	-------	------

6	Varun	21	CSE	92	Chennai
1	Anil	20	CSE	85	Hyderabad
3	Ravi	20	CSE	NULL	Hyderabad
2	Sneha	22	ECE	75	Mumbai
5	Kiran	22	ECE	75	Pune
9	Rekha	21	ECE	75	Pune
7	Priya	20	EEE	88	Delhi
10	Ramesh	23	EEE	NULL	Delhi
4	Divya	23	MECH	65	NULL
8	John	22	MECH	55	Mumbai

10 rows in set (0.00 sec)

mysql> select * from students order by age desc;

StudentID	Name	Age	Department	Marks	City
4	Divya	23	MECH	65	NULL
10	Ramesh	23	EEE	NULL	Delhi
2	Sneha	22	ECE	75	Mumbai
5	Kiran	22	ECE	75	Pune
8	John	22	MECH	55	Mumbai
6	Varun	21	CSE	92	Chennai
9	Rekha	21	ECE	75	Pune
1	Anil	20	CSE	85	Hyderabad
3	Ravi	20	CSE	NULL	Hyderabad
7	Priya	20	EEE	88	Delhi

10 rows in set (0.00 sec)

mysql> select * from students order by city,name;

StudentID	Name	Age	Department	Marks	City
4	Divya	23	MECH	65	NULL
6	Varun	21	CSE	92	Chennai
7	Priya	20	EEE	88	Delhi
10	Ramesh	23	EEE	NULL	Delhi
1	Anil	20	CSE	85	Hyderabad
3	Ravi	20	CSE	NULL	Hyderabad
8	John	22	MECH	55	Mumbai
2	Sneha	22	ECE	75	Mumbai
5	Kiran	22	ECE	75	Pune
9	Rekha	21	ECE	75	Pune

10 rows in set (0.00 sec)

mysql> select * from students order by marks desc,age asc;

StudentID	Name	Age	Department	Marks	City
-----------	------	-----	------------	-------	------

6	Varun	21	CSE	92	Chennai
7	Priya	20	EEE	88	Delhi
1	Anil	20	CSE	85	Hyderabad
9	Rekha	21	ECE	75	Pune
2	Sneha	22	ECE	75	Mumbai
5	Kiran	22	ECE	75	Pune
4	Divya	23	MECH	65	NULL
8	John	22	MECH	55	Mumbai
3	Ravi	20	CSE	NULL	Hyderabad
10	Ramesh	23	EEE	NULL	Delhi

10 rows in set (0.00 sec)

mysql> select * from students order by department,city,name;

StudentID	Name	Age	Department	Marks	City
6	Varun	21	CSE	92	Chennai
1	Anil	20	CSE	85	Hyderabad
3	Ravi	20	CSE	NULL	Hyderabad
2	Sneha	22	ECE	75	Mumbai
5	Kiran	22	ECE	75	Pune
9	Rekha	21	ECE	75	Pune
7	Priya	20	EEE	88	Delhi
10	Ramesh	23	EEE	NULL	Delhi
4	Divya	23	MECH	65	NULL
8	John	22	MECH	55	Mumbai

10 rows in set (0.00 sec)

mysql> /* Execution of order

/*> 1.from

/*> 2.Where

/*> 3.Group by

/*> 4.Having

/*> 5.Select

/*> 6.Order by */

mysql> select * from students;

StudentID	Name	Age	Department	Marks	City
1	Anil	20	CSE	85	Hyderabad
2	Sneha	22	ECE	75	Mumbai
3	Ravi	20	CSE	NULL	Hyderabad
4	Divya	23	MECH	65	NULL
5	Kiran	22	ECE	75	Pune
6	Varun	21	CSE	92	Chennai
7	Priya	20	EEE	88	Delhi
8	John	22	MECH	55	Mumbai

9	Rekha	21	ECE	75	Pune
10	Ramesh	23	EEE	NULL	Delhi

10 rows in set (0.00 sec)

```
mysql> select department,avg(marks) as avg_marks from students where age>20 group by department having avg(marks)>70 order by avg_marks desc;
```

department	avg_marks
CSE	92.0000
ECE	75.0000

2 rows in set (0.00 sec)

```
mysql> select city, count(*) as total_students from students where marks>70 group by city having count(*)>1 order by total_students desc;
```

city	total_students
Pune	2

1 row in set (0.00 sec)

```
mysql> select department, sum(marks) as total_marks from students where city in('Hyderbad','Pune') group by department having sum(marks)>100 order by total_marks desc;
```

department	total_marks
ECE	150

1 row in set (0.00 sec)

```
mysql> select city, max(marks) as highest_marks from students where department='ECE' group by city having max(marks)>70 order by highest_marks desc;
```

city	highest_marks
Mumbai	75
Pune	75

2 rows in set (0.00 sec)

```
mysql> select department, avg(age) as avg_age from students where city is not null group by department having avg(age)>20 order by department;
```

department	avg_age
CSE	20.3333
ECE	21.6667

	EEE		21.5000	
	MECH		22.0000	
+-----+-----+				

4 rows in set (0.00 sec)

mysql> select department, avg(age) as average_age from students where city is not null group by department having average_age>90 order by department;
Empty set (0.00 sec)